INDIAN STATISTICAL INSTITUTE

Parametric Inference

(B.Stat. 3rd year)

Semester 1, 2026–27

Homework 2

1. Prove that for observation $X \sim N(\theta, 1)$, X is a minimax estimate for $-\infty < \theta < \infty$.

2. Show that for observation $X \sim \text{Poisson}(\theta)$, any estimate for $0 < \theta < \infty$ has maximum risk ∞.

3. For $X_1, X_2, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Uniform}\left[\theta - \frac{1}{2}, \theta + \frac{1}{2}\right]$, compute the generalized Bayes estimate under improper prior $\pi(\theta) = 1 \;\; \forall\, \theta \in \mathbb{R}$.

4. Generate
$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \frac{1}{\pi\{1 + (x-\theta)^2\}}$$
for $\theta = 0$ and $n = 10, 15, 20$. Pretend $\theta \in \mathbb{R}$ is unknown. Compute MLE for θ based on your generated data.

5. Let $X \sim f(x)$ where
$$H_0 : f(x) = \tfrac{1}{2}e^{-|x|} \qquad \text{and} \qquad H_A : f(x) = N(0,1).$$
Derive most powerful test with level $0 < \alpha < 1$.